

ANKAN DERIA

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[in](#) [LinkedIn](#) [GitHub](#) [Portfolio](#) [Google Scholar](#)

RESEARCH INTERESTS

Researcher in multimodal AI working on **world models and long-horizon planning**, video generation, vision-language models, and unified multimodal architectures. **11 papers** (7 accepted, 4 under review; **8 first-author**) at venues including NeurIPS, CVPR, ACM-MM and AAAI. Currently an M.Sc. student in Computer Vision at MBZUAI, advised by Prof. **Salman Khan**.

EDUCATION

Mohamed bin Zayed University of Artificial Intelligence **Aug 2025 – Expected 2027**
*M.Sc. in Computer Vision; Advisor: Prof. **Salman Khan**; GPA: 3.7/4 (First Year)* *Abu Dhabi, UAE*

Government College of Engineering and Leather Technology **Aug 2020 – Jul 2024**
B.Tech. in Computer Science and Engineering; CGPA: 9.26/10 *Kolkata, India*

PUBLICATIONS

OmniSE: Support-Faithful Optimization for Evidence-Centric Audio-Video Reasoning

Authors: X. He, Y. Shen, **Ankan Deria**, X. Yue, M. H. Khan

Accepted — NeurIPS 2026

CoME-VL: Scaling Complementary Multi-Encoder Vision-Language Learning

Authors: **Ankan Deria**, K. Kumar, X. He, I. Razzak, H. Cholakal, F. S. Khan, S. Khan

Accepted — ACM-MM 2026 · Project Page

MedMO: Grounding and Understanding Multimodal Large Language Model for Medical Images

Authors: **Ankan Deria**, K. Kumar, A. M. Dukre, E. Segal, S. Khan, I. Razzak

Accepted — CVPR 2026 · Project Page

NeMoColor: Unified Noise Modulation for Enhanced Diffusion-based Image Colorization

Authors: **Ankan Deria**, D. Mahapatra, M. Mondal, S. Roy

Accepted — AAAI 2026 · GitHub

ViMaR: Dual-Stage Value-Guided Inference with Margin-Based Reward Adjustment for Fast and Faithful VLM Captioning

Authors: **Ankan Deria**, A. M. Dukre, F. Tang, S. Atito, S. Roy, M. Awais, M. H. Khan, I. Razzak

Accepted — NeurIPS 2025 · GitHub

Robust semantic learning for precise medical image segmentation

Authors: S. Chakraborty, K. Kumar, **Ankan Deria**, D. Mahapatra, B. Bozorgtabar, S. Roy

Accepted — Biomedical Signal Processing and Control, 2025 · GitHub

InVERGe: Intelligent Visual Encoder for Bridging Modalities in Report Generation

Authors: **Ankan Deria**, K. Kumar, S. Chakraborty, D. Mahapatra, S. Roy

Accepted — CVPRW 2024 · GitHub

WorldGuide: Goal-Directed Video World Model for Procedural Task Execution

Authors: **Ankan Deria**, K. Kumar, H. Cholakal, F. S. Khan, S. Khan

Under Review — ICLR 2027 · Project Page

Training-Free Speech-Centric Omni Understanding with Frozen VLMs

Authors: **Ankan Deria**, H. Rasheed, X. He, F. S. Khan, S. Khan

Under Review — AAAI 2027 · Project Page

MuGa-VTON: Multi-Garment Virtual Try-On via Diffusion Transformers with Prompt Customization

Authors: **Ankan Deria**, D. Mahapatra, B. Bozorgtabar, M. Chakraborty, S. Chakraborty, S. Roy
Under Review — TMLR 2026 · GitHub

SafeDiffusion-R1: Online Reward Steering for Safe Diffusion Post-Training

Authors: K. Kumar, **Ankan Deria**, A. Basu, F. Shamshad, H. Cholakkal, K. Nandakumar
Under Review — AAAI 2027 · Project Page

RESEARCH EXPERIENCE

Mohamed bin Zayed University of Artificial Intelligence

Aug 2025– Present

*M.Sc. Student, advised by Prof. **Salman Khan***

Abu Dhabi, UAE

- **WorldGuide**: goal-directed video world model for long-horizon procedural tasks (under review, ICLR 2027).
- **MedMO**: multimodal LLM for medical image grounding and understanding (CVPR 2026; Compute Transparency Champion).
- **CoME-VL**: complementary multi-encoder vision-language learning (ACM-MM 2026).

Mohamed bin Zayed University of Artificial Intelligence

Apr 2025– Aug 2025

*Research Assistant, advised by Prof. **Imran Razzak***

Abu Dhabi, UAE

- **ViMaR**: value-guided inference with margin-based rewards for fast, faithful VLM captioning (NeurIPS 2025).

Jio Institute

Jul 2024– Apr 2025

*Researcher, advised by Prof. **Sudipta Roy***

Navi Mumbai, India

- **NeMoColor**: noise modulation for diffusion-based image colorization (AAAI 2026).
- **MuGa-VTON**: diffusion-transformer multi-garment virtual try-on (under review, TMLR); medical image segmentation (BSPC 2025).

Jio Institute

Jul 2023– Oct 2023

*Research Intern, advised by Prof. **Sudipta Roy***

Navi Mumbai, India

- **InVERGe**: visual encoder for radiology report generation (CVPRW 2024).

LEADERSHIP

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Dec 2022– Aug 2023

Chapter Lead

Remote

- Led the Kolkata chapter's open-source NLP projects: **Text Summarization**, **Hate Speech Detection**, and **Job Recommendation System**.

HONORS AND AWARDS

- **CVPR 2026 Compute Transparency Champion Award** (for MedMO)
- **3rd place**, MBZUAI Research Showcase 2025
- **69th rank**, Amazon ML Challenge, 2023; selected for Amazon ML Summer School, 2022
- **Reviewer**: NeurIPS, ICLR, WACV, BMVC

SKILLS

- **Languages**: Python, $\text{\LaTeX}$
- **Frameworks**: PyTorch, Hugging Face Transformers, Diffusers, DeepSpeed, PEFT, Gradio
- **Infrastructure**: Large-scale distributed training (multi-GPU / multi-node), Docker, GCP, Vertex AI

REFERENCES

Prof. Salman Khan

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Prof. Fahad Khan

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Prof. Sudipta Roy

Associate Professor

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